A.3 (a) $\underline{\cup_{\alpha\in I}A_\alpha = (-1,2)}$

$\because \forall\alpha \in I = [0,1], A_\alpha = (\alpha-1,\alpha+1) \subset (-1,2)$, $\therefore \cup_{\alpha\in I}A_\alpha \subset (-1,2)$. On the other hand, $\forall x \in (-1,2), \exists\alpha \in I$, such that $x \in A_\alpha$, e.g. $\alpha = \max(x,0)/2$, $\therefore (-1,2) \subset \cup_{\alpha\in I}A_\alpha$.

$\underline{\cap_{\alpha\in I}A_\alpha = (0,1)}$

$\forall\alpha \in I, \alpha-1 \leq 0 < 1 \leq \alpha+1, \therefore (0,1) \subset \cap_{\alpha\in I}A_\alpha$.
Further $\cap_{\alpha\in I}A_\alpha \subset A_0 \cap A_1 = (0,1)$.

(b) $\underline{\cup_{\alpha\in J}A_\alpha = \cup_{\alpha\in I}A_\alpha = (-1,2)}$

$\cup_{\alpha\in J}A_\alpha \subset \cup_{\alpha\in I}A_\alpha$. $\mathbb{Q}$ is dense in $\mathbb{R} \Rightarrow \forall\alpha \in I, \exists p,q \in \mathbb{Q}\cap I$ such that $p \leq \alpha \leq q$. $A_\alpha \subset (p-1,q+1) = A_p \cup A_q \subset \cup_{\alpha\in J}A_\alpha$, $\therefore \cup_{\alpha\in I}A_\alpha \subset \cup_{\alpha\in J}A_\alpha$.

$\underline{\cap_{\alpha\in J}A_\alpha = \cap_{\alpha\in I}A_\alpha = (0,1)}$

$\cap_{\alpha\in I}A_\alpha \subset \cap_{\alpha\in J}A_\alpha \subset A_0 \cap A_1 = (0,1) = \cap_{\alpha\in I}A_\alpha$.

A.5 The notation $|\Omega|$ means the cardinality of Ω.

First of all, $|\mathcal{P}(\mathbb{N})| = |\{0,1\}^{\mathbb{N}}|$, because there exists a 1-1 (onto) map between them. For example, one candidate f would be:

$$f(A) = \{\omega_i\}_{i\in\mathbb{N}},\ \omega_i = I\{i \in A\}, \quad \forall A \in \mathcal{P}(\mathbb{N})$$

Secondly, $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$. $\{\{i\} : i \in \mathbb{N}\} \subset \mathcal{P}(\mathbb{N})$ reveals that $\mathcal{P}(\mathbb{N})$ is infinite. If $\mathcal{P}(\mathbb{N})$ is countable, then there exists a 1-1 (onto) map $g : \mathbb{N} \to \mathcal{P}(\mathbb{N})$. Let $A = \{i \in \mathbb{N} : i \notin g(i)\}$. Then $\exists x \in \mathbb{N}$ such that $g(x) = A$ since $A \in \mathcal{P}(\mathbb{N})$. Now $x \in A \Rightarrow x \notin g(x) = A$, and $x \notin A \Rightarrow x \notin g(x) \Rightarrow x \in A$, a contradiction.

The above proof remains valid for any set Ω, that is, $|\Omega| < |\mathcal{P}(\Omega)|$.

A.9a Note that $\sum_{j=1}^{1} j^2 = 1^2 = 1 = \frac{1\times2\times3}{6}$, i.e. the formula holds for $n=1$. Suppose the formula holds for $n=k$: $\sum_{j=1}^{k} j^2 = \frac{k\times(k+1)\times(2k+1)}{6}$ then for $n=k+1$, we have

$$\begin{aligned}
\sum_{j=1}^{(k+1)} j^2 &= \sum_{j=1}^{k} j^2 + (k+1)^2 = (k+1)\left[(k+1) + \frac{k\times(2k+1)}{6}\right] \\
&= (k+1)\left[\frac{(k+2)\times(2k+3)}{6}\right] \\
&= \frac{(k+1)\times((k+1)+1)\times(2(k+1)+3)}{6}
\end{aligned}$$

So, the formula holds for $n=k+1$, and hence for all $n \geq 1$.

A.9b (i) The formula is easily seen to be true when $n = 1$. Suppose it's true for $n = k$, that is

$$(x_1 + x_2)^k = \sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k-r}$$

Then

$$\begin{aligned}
(x_1 + x_2)^{k+1} &= (x_1 + x_2)^k (x_1 + x_2) \\
&= \sum_{r=0}^{k} \binom{k}{r} x_1^{r+1} x_2^{k-r} + \sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k+1-r} \\
&= \sum_{l=1}^{k+1} \binom{k}{l-1} x_1^l x_2^{k+1-l} + \sum_{r=0}^{k} \binom{k}{r} x_1^r x_2^{k+1-r} \\
&= x_2^{k+1} + \sum_{r=1}^{k} \left(\binom{k}{r-1} + \binom{k}{r} \right) x_1^r x_2^{k+1-r} + x_1^{k+1} \\
&= x_2^{k+1} + \sum_{r=1}^{k} \binom{k+1}{r} x_1^r x_2^{k+1-r} + x_1^{k+1} \\
&= \sum_{r=0}^{k+1} \binom{k+1}{r} x_1^r x_2^{k+1-r}
\end{aligned}$$

This finishes the proof.

(ii) Formula is true when $k = 1, 2$ for all $n \in \mathbb{N}$. Suppose it's true for $k = m, \forall n \in \mathbb{N}$,

$$(x_1 + \ldots + x_m)^n = \sum_{r_1 + \ldots + r_m = n} \frac{n!}{r_1! \cdots r_m!} x_1^{r_1} \cdots x_m^{r_m}$$

where $0 \leq r_i \leq n, r_i \in \mathbb{N}, i = 1, \ldots, m.$

Then for $k = m + 1$

$$\begin{aligned}
(x_1 + \ldots + x_{m+1})^n &= \sum_{s=0}^{n} \binom{n}{s} (x_1 + \ldots + x_m)^s x_{m+1}^{n-s} \\
&= \sum_{s=0}^{n} \sum_{r_1 + \ldots + r_m = s} \frac{n!}{r_1! \ldots r_m! (n-s)!} x_1^{r_1} \ldots x_m^{r_m} x_{m+1}^{n-s} \\
&= \sum_{r_1 + \ldots + r_{m+1} = s} \frac{n!}{r_1 \ldots r_{m+1}!} x_1^{r_1} \ldots x_{r_{m+1}}^{r_{m+1}}. \quad (r_{m+1} = n - s)
\end{aligned}$$

This finishes the proof.

A.11 $\forall a \in A \Rightarrow a \leq 2$, so A is bounded above in $\mathbb{Q}$. Let $b = \sup A$ (the existence of $l.u.b.$ is explained in P 579). $\forall a \in A, a \leq b \Rightarrow b \geq \sqrt{2}$ (we can always find an increasing sequence of rational numbers converge to $\sqrt{2}$). By definition of $l.u.b.$, $\forall n \in \mathbb{N}$, $\exists a_n \in A, s.t. \sqrt{2} > a_n > b - 1/n$, let $n \to \infty \Rightarrow b \leq \sqrt{2}$. Thus $b = \sqrt{2}$.

A.12 $\forall n \in \mathbb{N}$,

$$\inf_{k\geq n} x_k + \inf_{k\geq n} y_k \leq x_n + y_n \leq \sup_{k\geq n} x_k + \sup_{k\geq n} y_k$$

$$\inf_{k\geq n}(x_k+y_k) \leq \sup_{k\geq n}(x_k+y_k) \Rightarrow \liminf_{n\to\infty}(x_n+y_n) \leq \limsup_{n\to\infty}(x_n+y_n)$$

Take lim sup and lim inf on all sides in the first inequality, we get

$$\liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n \leq \liminf_{n\to\infty}(x_n+y_n) \leq \limsup_{n\to\infty}(x_n+y_n) \leq \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n$$

A.17 (a) $\because \frac{n}{n+1} \leq (n/(n+1))^{1/n} \leq 1$, $\therefore (\limsup_{n\to\infty}|a_n|^{1/n})^{-1} = 1$

(b) $|a_n|^{1/n} = n^{p/n} = e^{p\log n/n}$, and $0 \leq \log n/n \leq \log n/(\sqrt{n}\log\sqrt{n}) \leq 2/\sqrt{n}$, $\therefore (\limsup_{n\to\infty}|a_n|^{1/n})^{-1} = 1$

(c) By Stirling's Formula, $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, $\therefore (\limsup_{n\to\infty}|a_n|^{1/n})^{-1} = +\infty$. Hmm, you don't like Stirling's Formula?

$$(n!)^{1/n} = (n(n-1)\cdots[n/2]\cdots 1)^{1/n} \geq (n(n-1)\cdots[n/2])^{1/n} \geq ([n/2])^{1/2},\ n\geq 2.$$

A.18 (a) Let $f(x) = \frac{1}{1-x}$, by direct calculation, we see that the n-th derivative of f is

$$f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}},\ n\in\mathbb{N}.$$

So the Taylor expansion of f at 0 will be $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x|<1$. It can be easily seen that the radius of convergence for the power series $\sum_{n=0}^{\infty} x^n$ is 1, which means that this power series converges in $I=(-1,1)$.

(b) Let $g(x) = 1+x+x^2$, then $g'(2)=5, g''(2)=2, g^{(n)}(2)=0, \forall n\geq 3$. So $1+x+x^2 = g(2)+\frac{g'(2)}{1!}(x-2)+\frac{g''(2)}{2!}(x-2)^2 = (x-2)^2+5(x-2)+7$.

(c) (i) For any $j\geq 1$, the $j-th$ derivative of $f(x)$ when $x\neq 0$ is

$$f^{(j)}(x) = \sum_{k=0}^{j-1}\frac{a_{k,j}}{e^{x^{-2}}x^{3j-2k}},\quad |x|<1, x\neq 0$$

, where $a_{k,j}$ are constants.

$$\lim_{h\to 0}\left|\frac{f(h)-f(0)}{h}\right| = \lim_{|h|\to 0}\frac{e^{-1/h^2}}{|h|} = \lim_{s\to+\infty} se^{-s^2} = 0,\quad (s=1/|h|)$$

Thus f is differentiable at $x=0$ and $f'(0)=0$. Suppose $f(x)$ is n-th order differentiable at $x=0$, and $f^{(n)}(0)=0$, then

$$\lim_{h\to 0}\left|\frac{f^{(n)}(h)-f^{(n)}(0)}{h}\right| \leq \lim_{|h|\to 0}\sum_{k=0}^{n-1}\frac{a_{k,n}}{e^{1/h^2}|h|^{3n-2k+1}}$$

$$= \lim_{s\to+\infty}\sum_{k=0}^{n-1}\frac{a_{k,n}s^{3n-2k+1}}{e^{s^2}} = 0.$$

That is f is $(n+1)$-th order differentiable at $x=0$ and $f^{(n+1)}(0)=0$. By principle of induction, f is infinitely differentiable at 0 and $f^{(j)}(0)=0$, $j\in\mathbb{N}$.

(ii) The Taylor series will be identical to 0 always because $f^{(j)}(0) = 0, j \geq 0$, thus will not converge to f on $(-1, 1)$.

A.24c The three unit balls are as shown below.